

FaceSecure:

Mobile-Based Secure Offline Facial Recognition and Liveness Detection System for Remote Highway Operations

Abstract

India's national highway network employs thousands of personnel stationed at remote toll plazas, rest stops, and construction checkpoints where consistent cellular or internet connectivity is absent. Conventional identity verification systems that rely on cloud-based authentication are rendered inoperable in these environments, creating significant security vulnerabilities and administrative gaps. This paper proposes FaceSecure, a fully offline, mobile-first facial recognition and liveness detection system designed for Android devices operated by the National Highways Authority of India (NHAI).

FaceSecure employs MobileFaceNet—a compact Convolutional Neural Network architecture optimized for real-time face embedding generation on resource-constrained mobile hardware—combined with a two-layer liveness detection pipeline comprising passive texture analysis and active challenge-response verification. All biometric embeddings are stored locally using AES-256 encryption within an on-device SQLite database, ensuring no biometric data leaves the device. The proposed system targets recognition accuracy exceeding 95%, liveness detection accuracy above 93%, and end-to-end authentication latency under two seconds on mid-range Android hardware. FaceSecure directly addresses NHAI's operational requirements for identity security, spoofing resistance, and zero dependency on network infrastructure.

1. Introduction

The deployment of reliable identity verification infrastructure across India's 1,46,000+ kilometre national highway network presents a formidable logistical challenge. Personnel stationed at geographically isolated checkpoints, remote toll collection booths, and construction monitoring stations routinely operate in areas with intermittent or absent network connectivity [1]. Traditional authentication mechanisms—including PIN-based access, smart card readers, and cloud-synchronized biometric systems—fail to meet the twin demands of security and availability in these environments.

Facial recognition has emerged as a compelling modality for secure, frictionless identity verification owing to its non-contact nature and high discriminative capability [2]. However, standard face recognition systems are susceptible to presentation attacks, wherein an adversary uses a printed photograph, a video replay on a screen, or a three-dimensional mask to bypass authentication. The academic literature terms this class of vulnerability a spoofing attack, and considerable research effort has been directed toward countermeasures collectively referred to as Presentation Attack Detection (PAD) or liveness detection [3].

A further constraint in highway deployment is compute budget: on-device processing must execute efficiently on Android smartphones without dedicated neural processing units, ruling out large deep learning models such as ArcFace variants trained on full-resolution networks. MobileFaceNet [1] addresses this constraint through depthwise separable convolutions and a bottleneck architecture that achieves competitive accuracy at less than one megabyte of model weight, making it uniquely suited to the target hardware profile.

This proposal presents FaceSecure, an integrated Android application that combines (i) offline biometric enrolment and authentication via MobileFaceNet embeddings, (ii) passive liveness detection through skin texture and frequency-domain analysis, (iii) active liveness verification via geometric landmark tracking for blink and head-turn challenges, and (iv) encrypted local storage of biometric templates. The system requires no internet access at any stage of operation.

2. Problem Statement

NHAI requires a field-deployable identity authentication solution satisfying the following operational constraints:

Constraint	Current State	Required Capability
Network Dependency	Cloud-authenticated systems fail at remote sites	100% offline operation
Device Portability	Fixed biometric kiosks impractical in field	Android smartphone-based authentication
Spoofing Resistance	Photo/video replay attacks easily defeat 2D face matchers	Passive + active liveness detection
Data Privacy	Biometric data transmitted to central servers	Encrypted on-device storage only
Hardware Cost	Specialised biometric readers costly to procure and maintain	Runs on mid-range Android (≤ 200 MB RAM allocation)

Table 1. Operational constraints and FaceSecure target capabilities.

3. Literature Review

The intersection of lightweight face recognition and on-device anti-spoofing has attracted substantial research attention, particularly following the widespread adoption of mobile biometric authentication in consumer electronics. Table 2 summarises the most directly relevant prior work.

Reference	Method	Key Contribution	Limitation	Relevance
Chen et al. [1], 2018	MobileFaceNet (depthwise sep. conv.)	<1 MB model, 99.55% LFW accuracy	No anti-spoofing layer	Core recognition engine
Bhaskar et al. [2], 2024	FaceNet + LivenessNet pipeline	97.4% spoof detection on photograph attacks	Cloud inference in many deployments	Liveness architecture reference
Eberz et al. [3],	Continuous mobile	Passive monitoring	High continuous	Active

2018	authentication with liveness	via facial landmarks	monitoring overhead	challenge design
Li et al. [4], 2019	FaceCloseup distortion-based PAD	Screen replay detection via display artefacts	High computational complexity	Passive liveness component
George & Marcel [5], 2019	Deep pixel-wise binary supervision	State-of-art PAD, cross-dataset generalization	Requires RGB-D or IR sensor	Passive model baseline

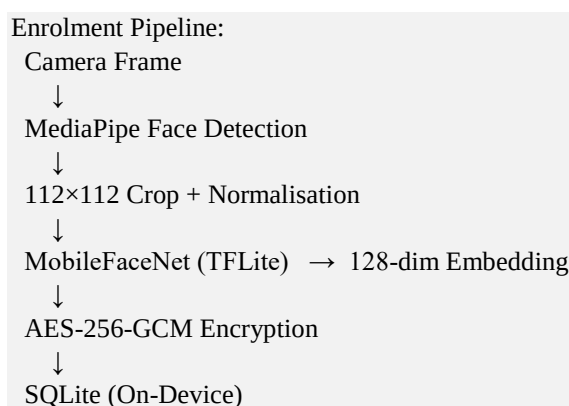
A consistent gap across reviewed works is the absence of a system combining lightweight on-device recognition, hybrid liveness detection (passive and active), and encrypted local storage in a single application targeting intermittent-connectivity environments. FaceSecure addresses this gap directly.

4. Proposed Methodology

4.1 Enrolment Phase

During enrolment, a NHAI administrator captures three to five reference photographs of an authorised individual. Each image undergoes the following pipeline:

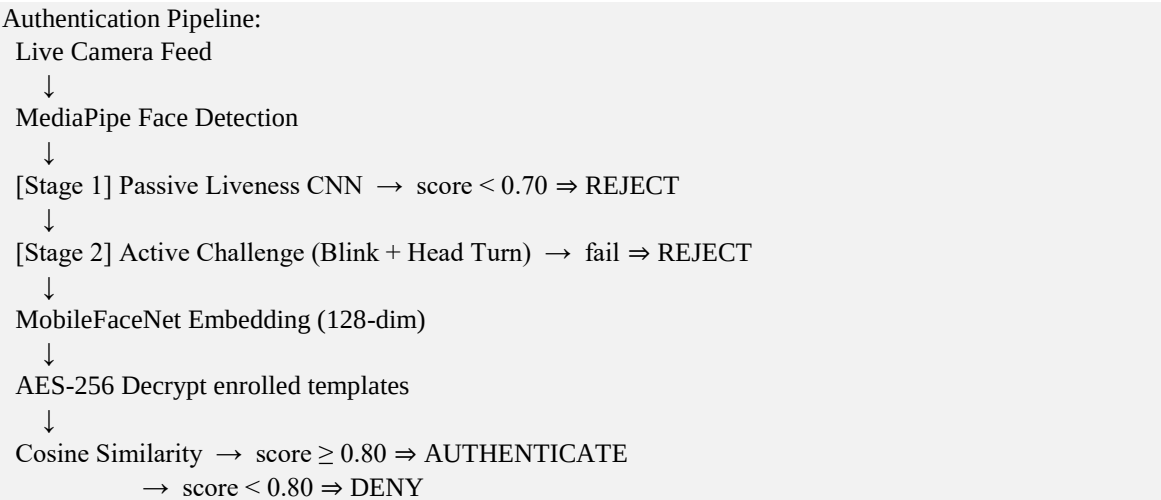
1. Face detection and alignment using MediaPipe Face Detection to extract a 112×112 normalised face crop.
2. Embedding generation: the cropped face is passed through the TensorFlow Lite MobileFaceNet model to produce a 128-dimensional embedding vector.
3. Template averaging: embeddings from multiple enrolment frames are averaged and L2-normalised to produce a single representative template.
4. Encryption: the template is serialised to bytes and encrypted using AES-256-GCM with a device-derived key stored in Android Keystore.
5. Persistence: the encrypted ciphertext and associated metadata are written to a local SQLite database with no plaintext biometric data at rest.



4.2 Authentication Phase

At authentication time, the live camera feed is processed through a two-stage liveness pipeline before face matching is attempted, ensuring that only a genuine live face can trigger a cosine similarity comparison:

- 6. Passive liveness check: a lightweight CNN binary classifier analyses skin texture frequency patterns, reflection artefacts, and colour distribution to distinguish a live face from a printed photograph or screen replay [4]. A confidence score below 0.70 terminates authentication immediately.
- 7. Active liveness challenge: MediaPipe Face Mesh extracts 468 facial landmarks. Eye Aspect Ratio (EAR) is computed across consecutive frames to detect a genuine blink; head pose is estimated from six canonical landmarks to verify a prompted head turn. The challenge sequence is randomised per session to prevent replay attacks.
- 8. Embedding extraction: only after both liveness stages pass is MobileFaceNet invoked to generate the query embedding.
- 9. Cosine similarity matching: the query embedding is compared against all enrolled templates. A similarity score ≥ 0.80 (empirically tuned threshold) returns an authenticated result; below this value, access is denied.
- 10. Audit logging: each authentication event—successful or failed—is appended to an encrypted local log with timestamp and employee ID for subsequent offline audit synchronisation.



5. System Architecture

FaceSecure is structured as a three-layer Android application:

Layer	Component	Technology
Presentation	Camera UI, enrolment wizard, challenge display, audit dashboard	React Native + Expo Camera
Intelligence	Face detection, embedding generation, passive/active liveness	TensorFlow Lite 2.x, MediaPipe Face Mesh, OpenCV 4
Data	Encrypted template storage, session log, admin config	SQLite, AES-256-GCM, Android Keystore

The application targets Android 10+ (API Level 29) and requires a minimum of 3 GB RAM and a camera with autofocus. All model inference runs on-device; the application has no network

permissions declared in the Android Manifest, enforcing air-gap compliance at the operating system level.

6. Novelty and Differentiation

The majority of hackathon submissions addressing biometric authentication propose a straightforward face recognition pipeline with cloud matching. FaceSecure differentiates itself across five orthogonal dimensions:

Dimension	Typical Competing Solution	FaceSecure Approach
Network Dependency	Matching performed on remote server	100% on-device — zero network calls
Liveness Strategy	Single passive check or none	Hybrid: passive texture CNN + active EAR/head-pose challenge
Model Weight	FaceNet ~80 MB, ArcFace ~100 MB	MobileFaceNet <1 MB TFLite
Storage Security	Embeddings in plaintext local DB	AES-256-GCM, key in Android Keystore
Hardware Target	High-end flagship or dedicated kiosk	Mid-range Android (≥ 3 GB RAM, any SoC)

7. Expected Results

Performance targets are derived from published benchmarks for MobileFaceNet on the Labeled Faces in the Wild (LFW) dataset [1] and anti-spoofing evaluations on the Replay-Attack and MSU-MFSD corpora [3], adjusted conservatively for operational variability in outdoor highway lighting conditions.

Performance Metric	Target Value	Justification
Face Recognition Accuracy (LFW protocol)	> 95%	MobileFaceNet baseline: 99.55% [1]; conservative outdoor derating
Passive Liveness Detection Accuracy	> 93%	LivenessNet ACER < 7% on Replay-Attack corpus [2]
Active Challenge Pass Rate (genuine users)	> 98%	EAR-based blink detection, 60 fps camera
End-to-End Authentication Latency	< 2 seconds	TFLite inference < 50 ms on Snapdragon 680 [1]
System Availability (offline)	100%	No network dependencies in Manifest
Peak RAM Consumption	< 200 MB	MobileFaceNet TFLite < 4 MB; OpenCV ~80 MB runtime
False Acceptance Rate (spoofing)	< 2%	Dual-stage liveness eliminates single-point spoof bypass

8. Future Scope

The FaceSecure architecture is intentionally modular to facilitate integration with existing government digital infrastructure:

- Aadhaar biometric bridging: an optional online synchronisation module could validate enrolled embeddings against the UIDAI offline face authentication API when connectivity is transiently available, enabling cross-site personnel transfer without physical re-enrolment.
- DigiLocker credential verification: integration with the DigiLocker API would allow verification of employee driving licences and service identity documents alongside face authentication, creating a multi-factor verification event.
- Edge server synchronisation: audit logs and anomaly events can be batch-uploaded to a regional NHAI edge server during scheduled connectivity windows, enabling centralised compliance reporting without real-time connectivity requirements.
- Thermal/NIR liveness extension: a supplementary infrared or depth sensor module would extend passive liveness robustness to silicone mask attacks, appropriate for higher-security checkpoint deployments.